

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**CourseCode:23CSE505****Credits:02****L:T:P:E2:0:2:2****ContactHours:60****Prerequisite(s):** Data Structures, Python Programming**Course Objectives****The objective of the course is to**

- Understand the basic concepts of Artificial Intelligence, terminologies related to AI in present business world.
- Learn the applications of search algorithms in Artificial Intelligence.
- Gain the knowledge representation in Artificial Intelligence.
- Acquire the knowledge of planning and reasoning in Artificial Intelligence.
- Use the different applications and techniques involved in AI for real time scenarios.

CourseOutcomes**At the end of the course, students will be able to**

Course Outcomes	Description	Bloom's Taxonomy Level
CO1	Outline the basics, importance and application of Artificial Intelligence.	Understanding(2)
CO2	Demonstrate the concept of rationality and various artificial agents with environments.	Applying(3)
CO3	Identify the suitable artificial agents in a problem-solving environment.	Understanding(2)
CO4	Employ the various searching techniques in AI domain.	Applying(3)
CO5	Outline the application of knowledge representation and reasoning in Artificial Intelligence.	Understanding(2)
CO6	Employ the planning techniques to understand the importance of Artificial Intelligence in real time environment.	Applying(3)

Course Contents

Module 1

[12hours]

Introduction to AI:

What is AI? Thinking humanly, Acting rationally. The Foundations of Artificial Intelligence, The History of Artificial Intelligence: The gestation of artificial intelligence, AI becomes an industry, Knowledge-based systems, The return of neural networks. The State of the Art, Risks and Benefits of AI.

Module2

[12hours]

Intelligent Agents:

Agents and Environments, Good Behavior: The concept of Rationality, How Agents Should Act, Structure of Intelligent Agents, Simple reflex agents, Goal-based agents, Utility-based agents, Nature of Environments.

Module3

[12hours]

Problem solving:

Problem-Solving Agents: Search problems and solutions, Formulating Problems, Well-defined problems and solutions, measuring problem-solving performance, Toy problems, Example Problems, search algorithms, uninformed search strategies, informed (Heuristic) search strategies, heuristic functions. Constraint satisfaction Problems: Definition, structure and examples.

Module-4

[12hours]

Knowledge representation and Reasoning:

Logical Agents: Knowledge based agents, The WUMPUS world, Logic, propositional logic, First order logic: Representation, syntax and semantics, using FOL, Knowledge Engineering in First order logic. Inference in FOL: Propositional vs First Order inference, Unification, forward chaining, Backward Chaining, Resolution

Module5

[12hours]

Planning:

Definition of classical planning, Algorithms for classical planning, Heuristics for planning, Planning and acting in non-deterministic domains. Time, schedules and resources, analysis of planning approaches.

TEXTBOOKS

1. Stuart J. Russell and Peter Norvig, “**Artificial Intelligence, A Modern Approach**”, Pearson Publication, 4th Edition, 2020.
2. Elaine Rich; Kevin Knight; Shivashankar BNair, “**Artificial Intelligence**”, New York McGraw Hill publication, Third edition, 2019

REFERENCES

1. D. Poole, “**Artificial Intelligence: Foundations of Computational Agents**”, Cambridge University Press, 2010.
2. Padhy N.P.: “**Artificial Intelligence and Intelligent Systems**”, 4th impression, Oxford University Press, 2007.
3. Nick Bostrom, “**Superintelligence paths, Dangers and Strategies**”, Kindle Edition, Oxford University Press, 2016.

Lab Course Contents:

The following programs can be implemented in python.

Lab Exp#	Details
1	Demonstrate the working principle of the following intelligent agents. a. Simple reflex agent b. Goal based agent
2	Implement the uninformed search algorithms a. Breadth First Search b. Depth First search
3	Implement the informed search algorithms a. A* b. AO*
4	Implement the Constraint Satisfaction Problem.
5	Implement the simple planning agent to solve problem. Example: A newspaper agent daily drops the newspaper to the area assigned in such a manner that he has to cover all the houses in the respective area With minimum travel cost. Compute the minimum travel cost. [travelling salespersons Problem]

Text book:

1. David L. Poole and Alan K. Mackworth, “**Artificial Intelligence: Foundations of Computational Agents**”, 2nd Edition, Cambridge University Press, 2017.

Online References: <https://artint.in/fo/AIPython/>